

# The empirical recurrence for OEIS A249713, proved from an exact model

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## Abstract

OEIS A249713 carries an empirical recurrence of order 8. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by an exact polynomial in  $n$ , from which  $a$  provably satisfies a MONIC linear recurrence of order  $S = 11$ , derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates  $a$  is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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## 1 The conjecture

OEIS A249713 is “Number of length  $7+3 \cdot n$  arrays with every four consecutive terms having the maximum of some two terms equal to the minimum of the remaining two terms.”. It has offset 1 and begins

72, 1177, 7056, 27113, 80360, 200489, 442144, 888465, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical:  $a(n) = (1/70)n^7 + (19/15)n^6 + (178/15)n^5 + 30n^4 + (851/30)n^3 + (56/15)n^2 - (446/105)n + 1$ .

As of the “Last modified” line on the live entry (Nov 10 2018 05:46:41, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

## 2 The count is a polynomial

The array has a FIXED length  $L$  and an alphabet  $0..n$  that grows. The condition is decided by the ORDER of the terms alone — by which are equal and which is larger — so an admissible array is described completely by its weak ordering: the ordered set partition of the  $L$  positions into  $m$  blocks of equal value, blocks listed in increasing order. Every assignment of  $m$  distinct values from  $\{0, \dots, n\}$  to those blocks, in increasing order, gives one array, and there are  $\binom{n+1}{m}$  of them. Hence

$$a(n) = \sum_{m=1}^L N_m \binom{n+1}{m},$$

with  $N_m$  the number of admissible weak orderings into  $m$  blocks, computed exactly.

### 3 The bound

That expression is a polynomial in  $n$  of degree at most  $L$ , so  $(z-1)^{L+1}$  annihilates  $a$  and  $S = 11$ . Nothing is fitted and no threshold is guessed: the  $N_m$  are counted, not estimated.

### 4 The proof

**Lemma 1.** *Let  $A(z) = z^S - \sum_{j < S} \alpha_j z^j$  be the monic annihilator derived above and  $q(z) = z^8 - \sum_i c_i z^{8-i}$  the characteristic polynomial of the conjectured recurrence. The residual  $r(n) = a(n) - \sum_i c_i a(n-i)$  is a linear combination of shifts of  $a$ , so  $A$  annihilates  $r$  as well. Since  $A(0) \neq 0$  the recurrence it gives runs in both directions, and  $S$  consecutive zeros of  $r$  force  $r$  to vanish at every later index.*

*Proof.*  $A$  annihilates  $a$ , hence every shift of  $a$ , hence any linear combination of shifts; that is  $r$ . A monic recurrence with nonzero constant term determines each term from the  $S$  before it and each from the  $S$  after it, so a run of  $S$  zeros propagates.  $\square$

**Theorem 2.**

$$a(n) = 8a(n-1) - 28a(n-2) + 56a(n-3) - 70a(n-4) + 56a(n-5) - 28a(n-6) + 8a(n-7) - a(n-8)$$

for every  $n > 0$ .

*Proof.* The terms  $a(0), \dots$  were computed exactly in integer arithmetic from the model, extended by  $A$  where the model itself was not evaluated, and  $r(n)$  formed for each index. Those integers vanish from  $n = 0 + 1$  onwards, and the run exceeds  $S + 8$ , which by Lemma 1 settles every larger index at once.  $\square$

### 5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 26 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator  $A$  was itself tested on terms it had not been given: the model was evaluated beyond the  $S$  values that determine it and  $A$  reproduced them. A bound that is too small fails this test, which is why it is run.

### References

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- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.
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